

DATA MANAGEMENT SYSTEMS DESIGN

PROJECT

Purpose of this project

Analyze, design, implement, and document a database system application. You will use the methodology for database development learned in class. The system must be implemented using a DBMS with any language as a host language for the application. The system must include the basic functionality described below.

THE RESEARCH LAB MANAGER

The following requirements are provided as a starting point to guide your system design. As the designer, you must analyze the mini-world and determine any additional details or features required for your implementation. Consequently, individual group designs and implementation styles may vary.

Data and Functional Requirements

This project aims to develop a comprehensive database system to support and streamline operations within a university research laboratory. The system will facilitate tracking of lab members, research projects, equipment usage, publications, and funding (provided by grants).

The lab includes faculty, students, and external collaborators. Each member participates in projects, may use shared equipment, and may contribute to publications. Projects are led by faculty. The lab also maintains a mentorship program among its members. Projects are funded by grants, which include budgets and reporting requirements.

Lab members have a **member ID**, **name**, **member type** (a lab member can only be a faculty member, a **student**, or an **external collaborator**), and a **join date**. For faculty, the system also records their **department**. Students have a **student number**, **academic level** (e.g., junior, senior, graduate student), and **major**. For collaborators, the system stores their institutional **affiliation** and a short **curriculum vitae**.

Projects have a **title**, a unique project **ID**, a **start date**, an **end date**, a **planned duration**, and a **status** (active, completed, or paused). Grants have a unique **ID**, a **funding agency**, a **budget**, a **start date**, and an expected **duration**. Each grant is issued to fund one project, although some projects may not be funded by any grant.

Each lab member works on at least one project, and their role in the project as well as hours of weekly involvement are recorded. A faculty member may lead multiple projects, but each project has exactly one leader.

Each equipment item has a unique **equipment ID**, **name**, **type**, and **manual**. For each equipment item, there are one or more devices, which represent the physical units of that equipment. Each device has a **purchase date**, a **status** (available, in use, or retired), and a **device number**. The device number uniquely identifies a device *within* its equipment, but devices belonging to different equipment items may share the same device number.

Devices may be used by lab members, and the system must record the usage **start date**, **end date**, and **purpose** of use. A device cannot be used by more than three members simultaneously.

Publications have a **unique ID**, a **title**, a **publication date** (including month and year), a **venue**, and an **optional DOI** (Digital Object Identifier).

Every publication recorded in the database is authored by at least one lab member, and typically multiple lab members collaborate in authoring a publication.

Lab members may mentor other lab members, and the start and end dates of each mentorship must be recorded. A mentee cannot have more than one mentor, and **students cannot mentor** faculty members.

Note that some constraints or property values may not be directly represented in the E/R diagram. However, the data structures must be designed to store these values and to allow enforcing these constraints.

The system must support queries, updates, and reporting for lab administration.

Each group can make further assumptions but: (a) they have to be reasonable, (b) they should not be in conflict with the assumptions described above, and (c) they have to be clearly stated in your report.